

FAQ

- What is the correct way to start the robot?

Answer: After turning on the robot's power, wait until you hear a long beep from the buzzer; that means the startup was successful.

- Why does the robot need parameter configuration?

Answer: Because each user's WiFi environment and IP address are different, parameters need to be configured according to the actual situation.

- Why does the robot's buzzer keep beeping "beep—beep—beep"?

Answer: The robot beeps "beep—beep—beep" when the battery is low. At this point the robot cannot be controlled. Please turn off the robot's power and then charge the robot.

- When using ROS2 to control the robot and view its status, the data may lag, or a single command may not be executed after it is sent. Why?

Answer: This is caused by WiFi network congestion. The ESP32 and the ROS2 program on the virtual machine exchange data over WiFi, so if too many users are using the WiFi, the network quality will be affected, which in turn affects the data exchange between the ESP32 and the ROS2 program on the virtual machine. It is recommended to use a dedicated 2.4G band WiFi for testing. Note that the current ESP32 model is esp32-s3, which does not support 5G band WiFi.

- Why can't the robot be controlled using the ROS2 program?

Answer: Check whether someone is currently using a USB Host gamepad to control the robot. If so, stop using the USB Host gamepad, and then the ROS2 program can control the robot, because the USB Host gamepad has higher control priority than ROS2 topic control.

- If you cannot access the Dify backend by entering the virtual machine IP address,

Answer: You can restart the Dify script in the root directory: `sh shutdown_dify.sh`

```
yahboom@VM:~$ sh shutdown_dify.sh
✓ Container docker-nginx-1      Removed      11.4s
✓ Container docker-sandbox-1    Removed      10.8s
✓ Container docker-plugin_daemon-1 Removed      0.2s
✓ Container docker-worker_beat-1 Removed      3.8s
✓ Container docker-api_websocket-1 Removed      4.9s
✓ Container docker-weaviate-1    Removed      11.4s
✓ Container docker-ssrf_proxy-1  Removed      11.4s
✓ Container docker-worker-1      Removed      11.4s
✓ Container docker-api-1         Removed      4.0s
✓ Container docker-web-1         Removed      0.2s
✓ Container docker-db_postgres-1 Removed      0.2s
✓ Container docker-redis-1       Removed      0.2s
✓ Container docker-init_permissions-1 Removed      0.0s
✓ Network docker_ssrf_proxy_network Removed      0.1s
✓ Network docker_default         Removed      0.1s
yahboom@VM:~$ sh bringup_dify.sh
[+] up 15/15
✓ Network docker_default         Created      0.0s
✓ Network docker_ssrf_proxy_network Created      0.0s
✓ Container docker-db_postgres-1 Healthy      3.3s
✓ Container docker-sandbox-1     Started      1.0s
✓ Container docker-web-1         Started      1.3s
✓ Container docker-ssrf_proxy-1  Started      2.2s
✓ Container docker-redis-1       Started      1.0s
✓ Container docker-init_permissions-1 Exited      2.3s
✓ Container docker-weaviate-1    Started      1.5s
✓ Container docker-plugin_daemon-1 Started      6.2s
✓ Container docker-api_websocket-1 Started      4.2s
✓ Container docker-api-1         Started      3.9s
✓ Container docker-worker-1      Started      4.3s
✓ Container docker-worker_beat-1 Started      4.2s
✓ Container docker-nginx-1       Started      6.6s
yahboom@VM:~$
```

